

AI-based Fault Prediction for CT Scanners

Project ID: 26-1-D-31

Supervisor: Dr. Reuven Cohen

Co-Supervisor from Philips: Michael Bikman

Presenting: Anat Rutman & Iris Kanter

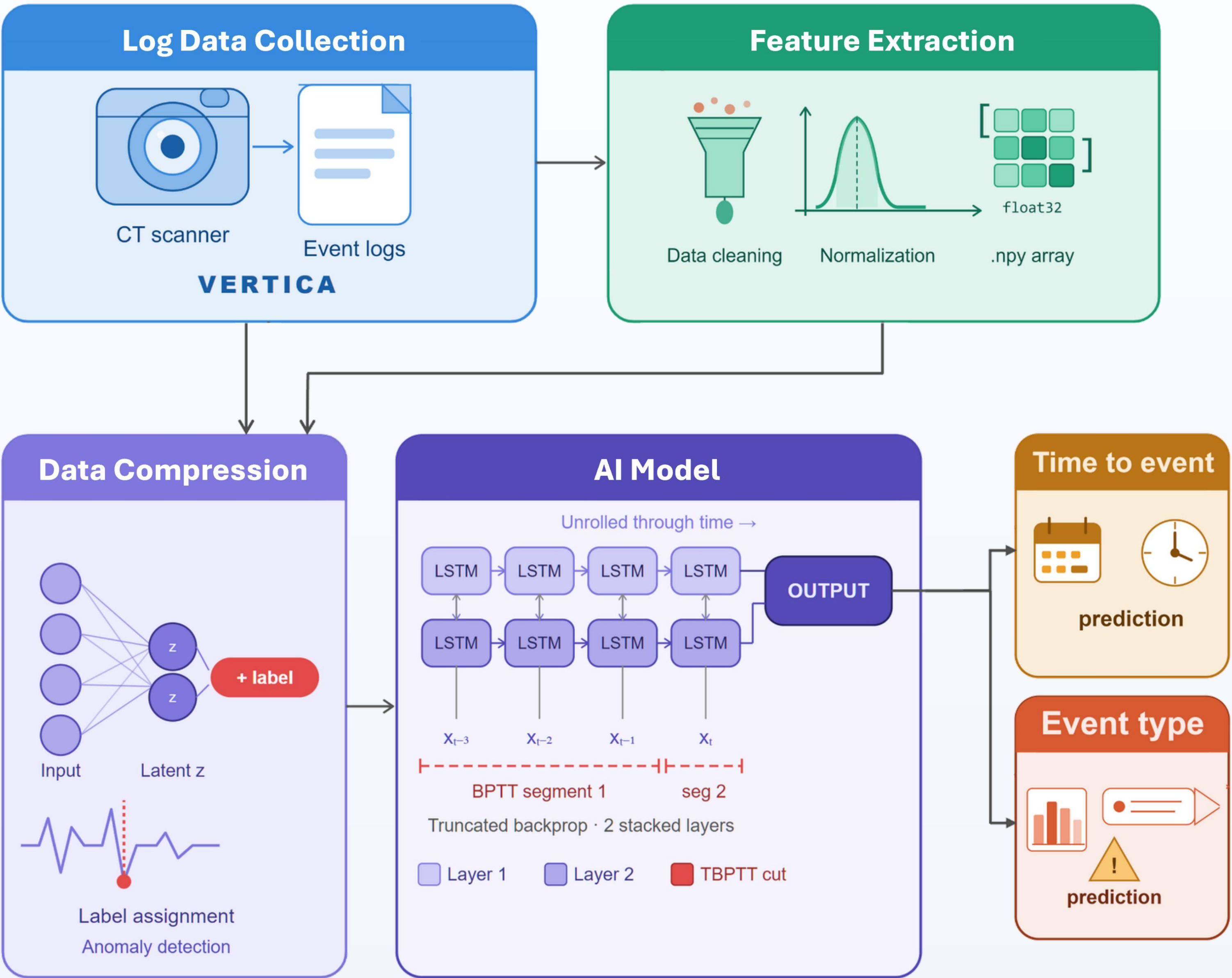
Project Goal

The goal is to predict upcoming scanner issues from historical log sequences.

CT scanners generate massive and complex operational logs, making it difficult to detect early signs of future faults manually.



System Architecture

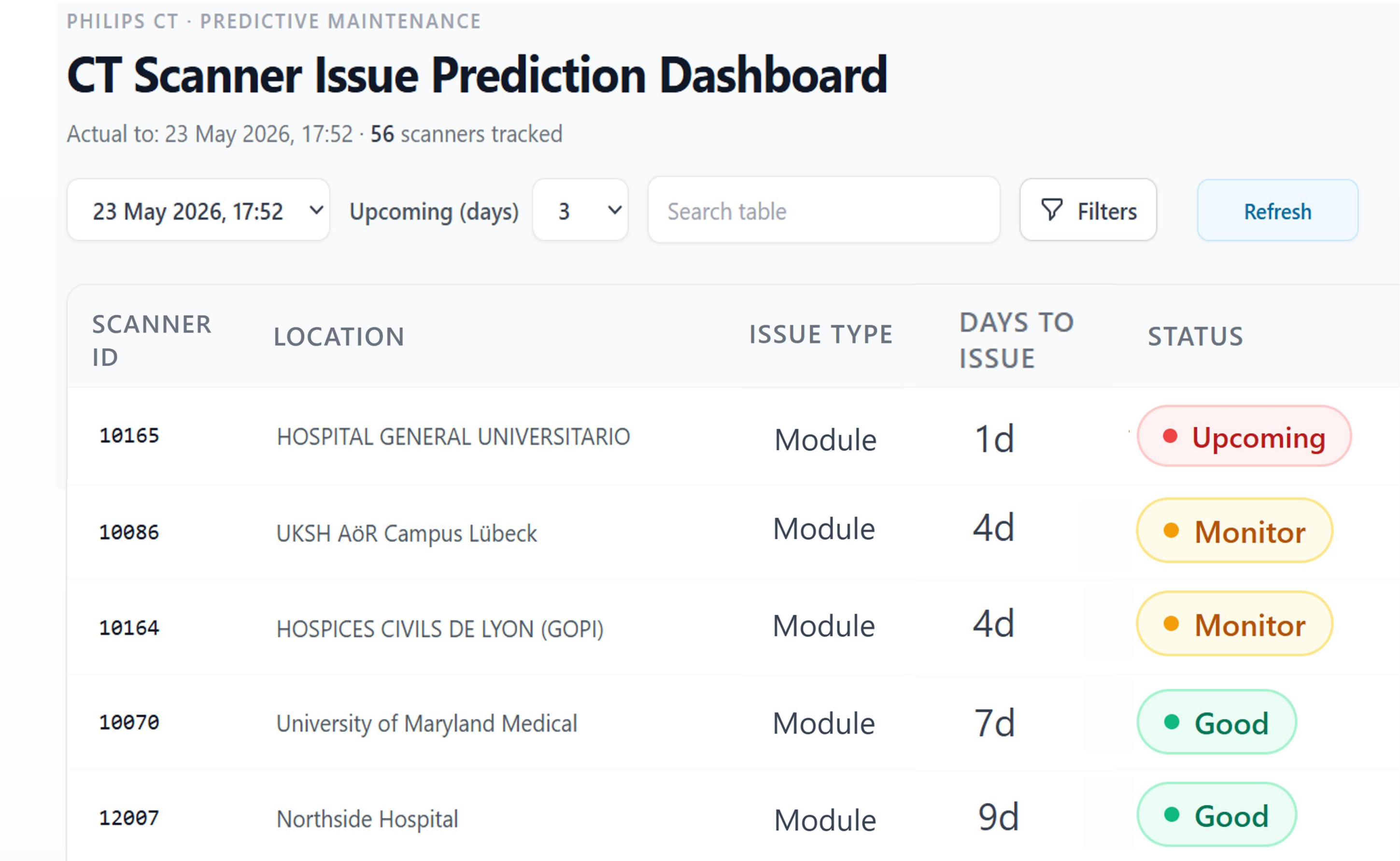


Results

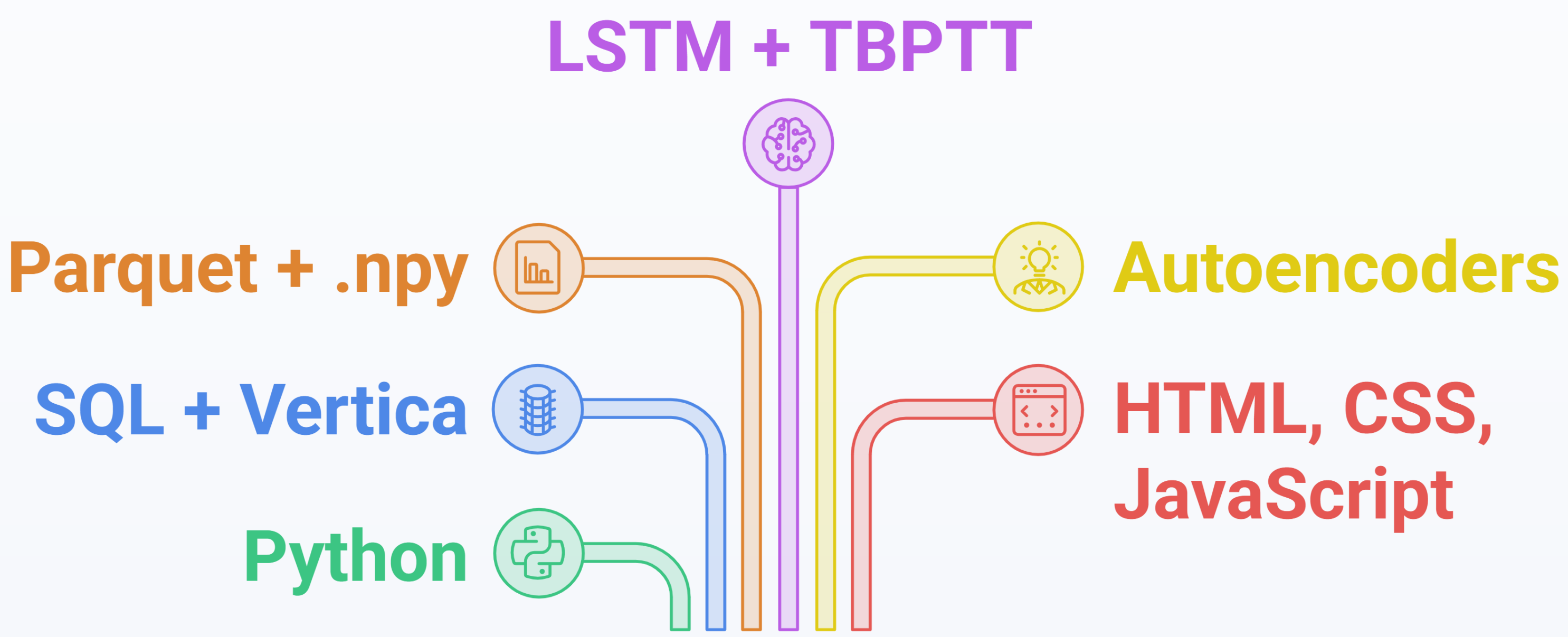
Metric	Raw-data model	Compressed-data model
Accuracy	80%	45%
Recall	30%	70%
Faults Found	4/13	9/13
False Positives	Low	Higher
Training Time	6 hours per epoch	30 minutes per epoch
Tuning & Experiments	not practical	easy
Main Finding	Misses many faults	Finds most real faults

Dataset covered:
18 log types, 60 scanners, 1 year of operation

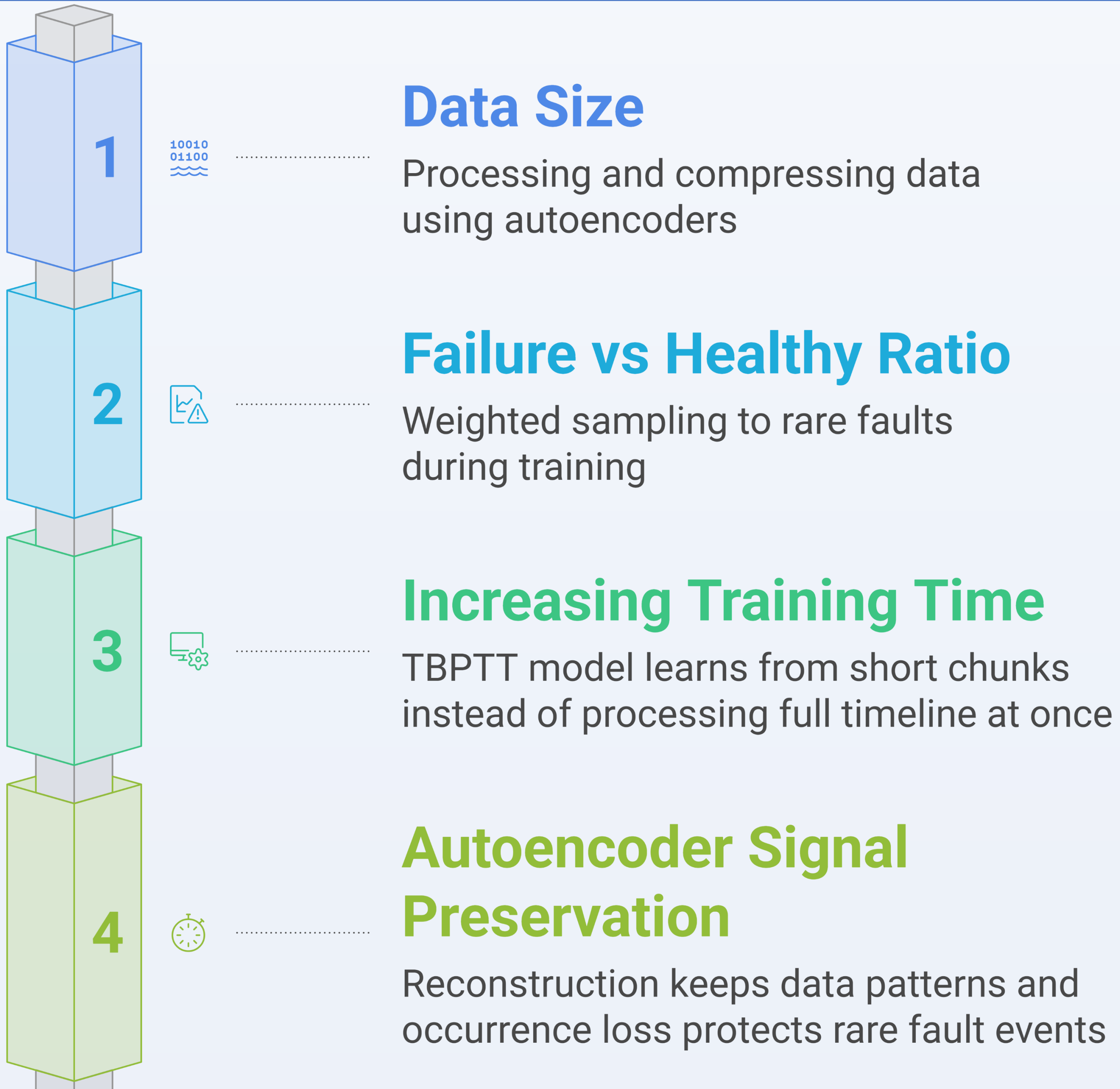
User Interface



Technologies Used



Challenges



Future Work

Model optimization and data filtering to improve accuracy and reduce false alarms.